Integration of e power x square in simple and easy way in details step by step

To solve the integral of e^(x²):

$$\int e^{x^2}\,dx$$

This is a famous integral because it cannot be solved using normal elementary functions (like polynomial, trig, log, etc.).

Step 1: Check substitution

Try:

$$u = x^2$$

Then:

$$du = 2x\,dx$$

But in the integral:

$$\int e^{x^2}\,dx$$

there is no x present, so substitution fails.

Step 2: Important Result

So:

$$\int e^{x^2}\,dx$$

has no simple closed-form answer.

It is written using a special function called Imaginary Error Function:

$$\int e^{x^2} dx = \frac{\sqrt{\pi}}{2} \operatorname{erfi}(x) + C$$

Step 3: If teacher asks "solve manually"

We use series expansion.

Since:

$$e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots$$

Put:

$$t = x^2$$

Then:

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \dots$$

Now integrate term by term:

$$\int e^{x^2} dx = \int \left(1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \dots\right) dx$$

So:

$$= x + \frac{x^3}{3} + \frac{x^5}{2 \cdot 5} + \frac{x^7}{6 \cdot 7} + \dots$$

Final Answer

Exact special-function form:

$$\int e^{x^2} dx = \frac{\sqrt{\pi}}{2} \operatorname{erfi}(x) + C$$

Series form:

$$x + \frac{x^3}{3} + \frac{x^5}{10} + \frac{x^7}{42} + ... + C$$

Easy Memory Trick

- $\int e^{-x^2} dx$ → error function
- $\int e^{x^2} dx$ → imaginary error function
- Both do not have simple normal answers.

If you meant Integration of e^{x^2} from limits (definite integral), then tell me because that can be solved numerically easily.

What is erfi and how it becomes square pi divide by 2 erfi(x)

Great question. You're asking where did this strange function come from:

$$\int e^{x^2} dx = \frac{\sqrt{\pi}}{2} \, \text{erfi}(x) + C$$

Let's explain in a simple easy way.

I. What is erfi(x)?

erfi means Imaginary Error Function.

It is a special mathematical function created because some integrals cannot be written using normal functions.

Just like:

- x^2, polynomial
- $\sin x$, trig function

- $\ln x$, log function

Some integrals need new named functions.

So mathematicians defined:

$$\operatorname{erfi}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{t^2} dt$$

This is the definition.

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2. Why was it created?

Because:

$$\int e^{x^2} dx$$

cannot be solved by:

- substitution
- integration by parts
- trig methods
- partial fractions

So instead of saying "no answer," mathematicians created a name:

$$\operatorname{erfi}(x)$$

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3. How does it become $\frac{\sqrt{\pi}}{2} \operatorname{erfi}(x)$?

Start from definition:

$$\operatorname{erfi}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{t^2} dt$$

Now multiply both sides by:

$$\frac{\sqrt{\pi}}{2}$$

Then:

$$\frac{\sqrt{\pi}}{2}\,\mathrm{erfi}(x) = \int_0^x e^{t^2}\,dt$$

So antiderivative becomes:

$$\int e^{x^2}\,dx = \frac{\sqrt{\pi}}{2}\,\mathrm{erfi}(x) + C$$

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4. Why √π specifically?

Because mathematicians chose this constant to make formulas clean and connected to Gaussian integrals like:

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

That's why π appears.

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5. Simple meaning

$$\mathrm{erfi}(x)$$

just means:

> "the accumulated area under e^{t^2} curve from 0 to x"

with a scaling factor.

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6. Easy analogy

Like:

- derivative of sin gives cos
- derivative of ln gives 1/x
- derivative of erfi gives exponential-square related term

Final Short Answer

$$\operatorname{erfi}(x)$$

is a special named function, defined as:

$$\operatorname{erfi}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{t^2}\, dt$$

So rearranging gives:

$$\int e^{x^2}\, dx = \frac{\sqrt{\pi}}{2} \operatorname{erfi}(x) + C$$

If you'd like, I can also ↪ explain why it is called "imaginary" error function (very interesting concept).

How we solve if power is minus x square

If the integral is:

$$\int e^{-x^2}\, dx$$

this is very famous and important.

I. Can normal integration methods solve it?

Try substitution:

$$u = -x^2$$

Then:

$$du = -2x\, dx$$

But there is no x in the integral, so normal substitution fails.

So:

$$\int e^{-x^2} dx$$

also has no elementary (simple standard) antiderivative.

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2. It is written using a special function: Error Function

Mathematicians define:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

So the antiderivative becomes:

$$\int e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \text{erf}(x) + C$$

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3. Why this one is more famous?

Because e^{-x^2} is the Gaussian curve used in:

- Probability / normal distribution
- Statistics
- Machine learning
- Physics
- Signal processing

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4. If teacher asks manual method (series)

Use expansion:

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \dots$$

Integrate term by term:

$$\int e^{-x^2} dx = x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \dots + C$$

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5. Important definite integral (very famous)

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

This is one of the most important results in mathematics.

Final Answer

$$\int e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}\,\mathrm{erf}(x) + C$$

where $\mathrm{erf}(x)$ is the error function.

Easy Memory Trick

- e^{-x^2} → erf(x)
- e^{x^2} → erfi(x)

Difference from previous one

Minus sign gives:

$$e^{-x^2}$$

which decays nicely, so it is used everywhere.

Plus sign gives:

$$e^{x^2}$$

which grows very fast.

If you'd like, I can also show how mathematicians prove $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$ step by step (beautiful trick).